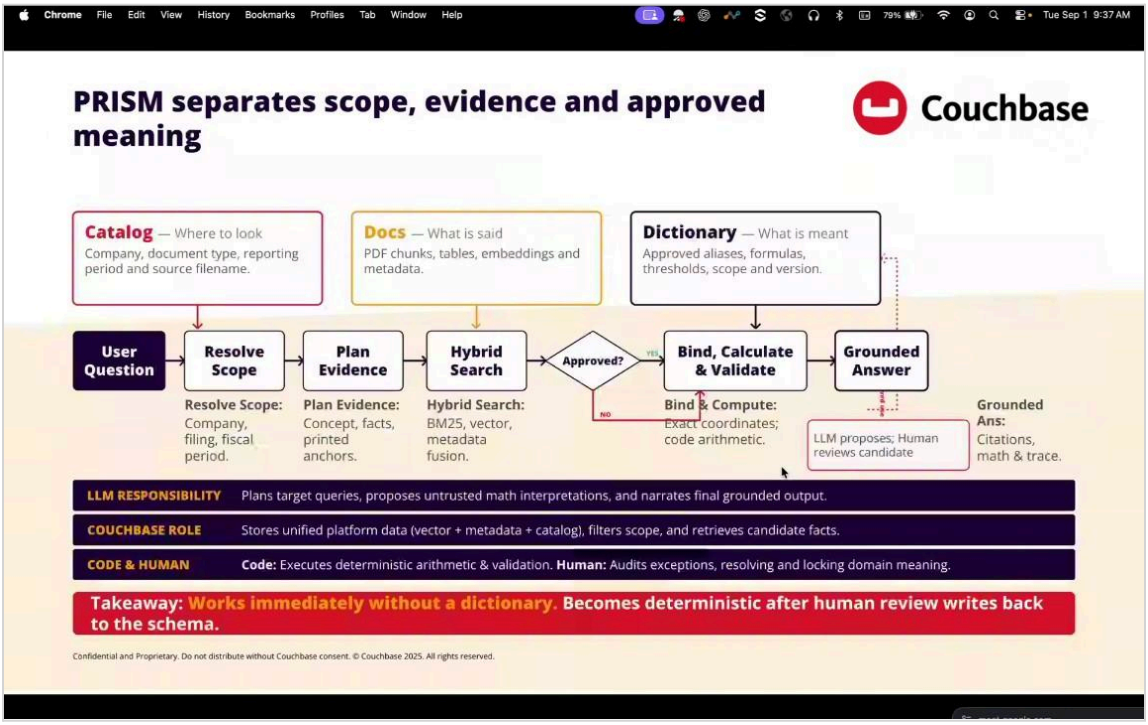
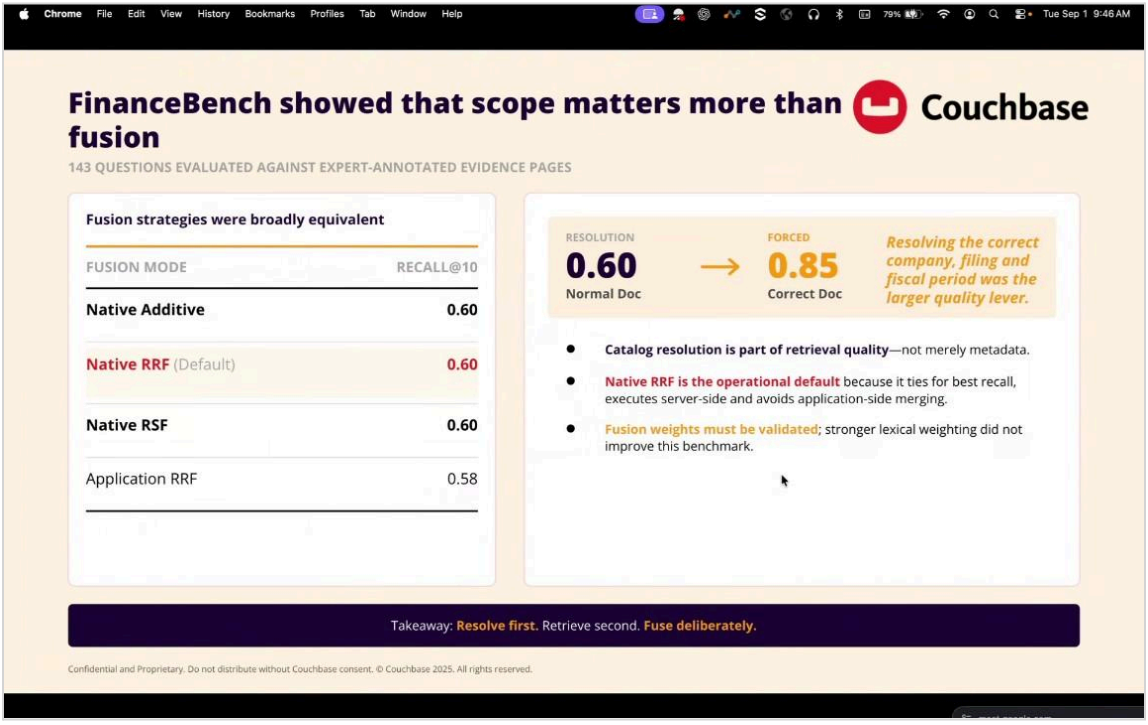
- **Prism Architecture - Challenges in Financial RAG**: Prasad Doddi presented "Prism," an architecture designed to solve limitations in standard Retrieval-Augmented Generation (RAG) for financial analysis (00:30:06) (00:34:22). Standard RAG fails with long SEC filings (100-500 pages) due to scope resolution issues (identifying the correct PDF), semantic failures (terms like "quick ratio" not being explicitly in text), and interpretation failures (requiring calculations not found in documents) (00:31:38). The architecture separates functionality into scope (Catalog), evidence (Chunks), and approved meaning (Dictionary/Ontology) (00:34:22).

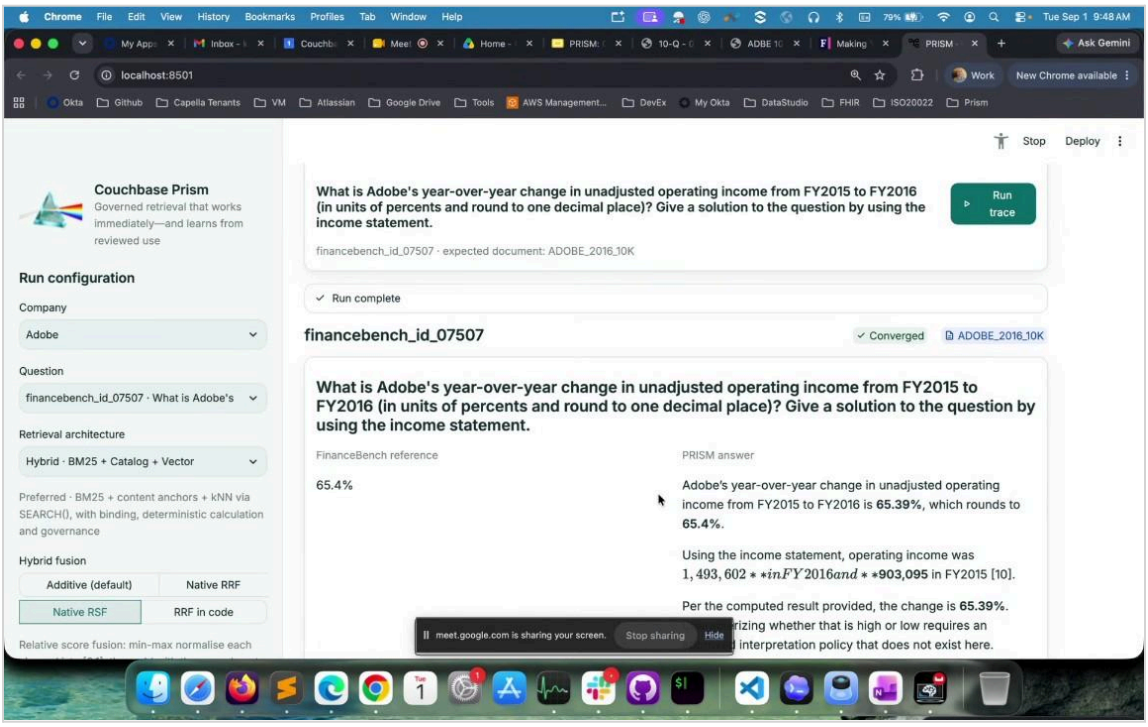


- **Prism Architecture - Runtime Workflow**: The runtime flow begins with the Large Language Model (LLM) planning evidence based on the user question and checking the catalog for scope (company, year, document type) (00:35:38) (00:38:20). The system performs a hybrid search—replacing vector-only search with a combination of BM25 lexical search and vector search—to retrieve grounded chunks. The LLM performs planning, binding, and narration, while the application handles deterministic mathematical calculations, ensuring the system remains governed and accurate (00:35:38) (00:39:21).
- **Prism Architecture - Benchmarking and Fusion Strategy**: Prasad Doddi utilized FinanceBench, an expert benchmark containing 368 PDFs and 150

expert questions with "gold" answers, to validate the Prism architecture (00:42:00). The system uses a fusion strategy—specifically Relative Score Fusion (RSF) or Reciprocal Rank Fusion (RRF)—which Couchbase search provides as a utility to balance lexical and semantic weights (00:40:43). This benchmark revealed that scope identification accounts for 50% of the success rate in RAG applications (00:43:27).



- **Prism Architecture - Demo and Learning Capabilities**: A live demonstration using Adobe and 3M financial data showed the system successfully identifying year-over-year changes in operating income and analyzing liquidity ratios (00:45:44) (00:51:01). The system learns "tribal knowledge" by allowing users to approve formulas, which are then saved into a dictionary for future use (00:53:14). Prasad Doddi noted that while the current implementation focuses on PDF documents, the architecture is compatible with any data plane supported by the AI data plane for chunking (00:55:45).



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